FARSuN Project Overview

Table of contents

Findability and Accessibility of historical Raw Sunspot Numbers

The International Sunspot Number (ISN) series, a cornerstone for space climate studies, still exhibits scale discrepancies despite recent recalibrations, underscoring the critical need for a comprehensive reconstruction from raw historical observations. The FARSuN project addresses this by systematically gathering, digitizing, and standardizing scattered sunspot data from 1607-1980, aiming to bridge the technical and scientific gap preventing holistic data use. Significant progress includes the digitization of Prof. Wolf's Mittheilungen (1610-1918) and the near-completion (95%) of more than 2000 Zürich observation tables, transforming them into machine-readable formats. Data extractions for challenging sources like Gruithuisen, Stark, Adams, and Rost are actively underway, supported by advanced tools like Transkribus and rigorous quality control efforts to define homogeneous metadata criteria. Upcoming initiatives include a citizen science web interface to engage the public in data recovery and the ultimate goal of constructing a new, consolidated ISN (V3) and historical hemispheric sunspot numbers. This comprehensive reconstruction will provide the Space Climate Community with an unprecedentedly accurate and accessible record, significantly improving solar cycle predictions, understanding long-term solar variability, space weather discussions, and refining climate models.

Dataset	**Description**	**Time Period Covered**	**Importance**	**Progress in Data Retrieval**	**Future Plans**
Zürich Tables	A collection of sunspot observations from various astronomers, primarily at the ETH Zurich.	1945 - 1979	Provides systematic records of sunspot activity, essential for solar variability studies.	Approximately 95% of over 2000 tables digitized; quality control processes are ongoing.	Complete digitization, integration into historical database, and public accessibility improvements.
Mittheilungen	Historical sunspot observations recorded by Rudolf Wolf and others at the Zürich Observatory.	1610 - 1918	One of the longest continuous records of sunspot observations, crucial for understanding long-term solar variability.	Digitized between 2017 and 2019; ongoing quality control and accessibility improvements.	Further validation, integration into broader historical database, and citizen science initiatives.
C.H. Adams	Sunspot observations made by C.H. Adams, including counts and descriptions of sunspot groups.	1819 - 1823	Provides detailed records of solar activity during a critical period, enhancing understanding of solar variability.	338 drawings extracted; ongoing quality control and cross-referencing with other datasets.	Continued validation, integration into historical database, and potential publication of findings.
Gruithuisen	Historical sunspot observations by Franz von Paula Gruithuisen, including weather conditions.	1817 - 1848	Offers insights into solar activity and its correlation with weather phenomena, contributing to climate studies.	Data extraction initiated; OCR tools being used for digitization and quality control.	Complete data extraction, integration into historical database, and public engagement initiatives.

Dataset	Description	Time Period Covered	Importance	Progress in Data Retrieval	Future Plans
Augustin Stark	Sunspot observations made by Augustin Stark, including counts and descriptions.	1813 - 1835	Important for understanding solar variability during the early 19th century.	Ongoing efforts to extract and digitize data; quality control measures being implemented.	Continued validation, integration into historical database, and potential publication of findings.